

Build an IR Light Emitter for IR Receivers

Introduction

Imagine you want to build an **infrared beam barrier** to detect when an object crosses a path. Naturally, you need two circuits: an **IR transmitter** and an **IR receiver**.

For the receiver, the first component that probably comes to mind is a **photodiode**. And technically, it can detect infrared light.

But there is a problem.

A simple photodiode cannot distinguish between the IR signal from your transmitter and the infrared radiation coming from the **sun, lamps, and other sources**. Your system could therefore produce false detections.

So, how can we make the receiver detect **only our IR signal**?

This is where **integrated infrared receivers with built-in filtering** come in.

These receivers combine a **photodiode, amplification, filtering, and signal processing** in a single package. Their internal filtering is designed to detect infrared signals modulated at a specific carrier frequency, making them much less sensitive to continuous or unwanted infrared sources such as sunlight and ambient lighting.

Different receivers are designed for different carrier frequencies. For example, **TSOP36** receivers are designed for approximately **36 kHz**, while **TSOP38** receivers are designed for approximately **38 kHz**. Other variants are available for frequencies such as **30 kHz, 40 kHz, or 56 kHz**.

This also imposes a requirement on our transmitter: we cannot simply turn the IR LED on and off. We need to **modulate the emitted infrared light at the carrier frequency expected by the receiver**.

In our case, since we are using a **38 kHz IR receiver**, the transmitter must generate a **38 kHz carrier**.

In this project, we will therefore build a simple **38 kHz IR emitter** designed to work with 38 kHz IR receivers such as the TSOP38.

What You Will Learn

By the end of this tutorial, you will be able to:

- Understand how an IR emitter built around two astable oscillators works.
- Calculate and select the component values according to the type of IR receiver being used.
- Build and test the complete IR emitter circuit.

Step 1: Understand the Circuit

This IR emitter is built around **two NE555 astable oscillators**.

The first oscillator runs at a low frequency of approximately **10 Hz** and controls the second oscillator, which generates the **38 kHz carrier signal**.

The second NE555 is controlled through its **RESET pin**. It only generates the 38 kHz signal when the output of the first oscillator is HIGH. As a result, the IR LED does not emit a continuous 38 kHz signal, but rather **short bursts of 38 kHz pulses separated by pauses**.

These pauses are important because the TSOP38 is designed to detect **bursts of modulated IR rather than a continuous 38 kHz signal**. Continuous transmission can cause its **automatic gain control (AGC) and interference suppression mechanisms** to reduce or reject the signal, preventing the receiver from responding correctly.

The first oscillator therefore acts as a **burst generator**, while the second oscillator generates the **38 kHz carrier** used by the IR receiver.

Step 2: Dimension the Two Astable Oscillators

2.1 – 10 Hz Astable Oscillator

This oscillator is built around **two resistors (R1 and R2) and a capacitor (C)**. This RC network determines the oscillation frequency.

The period is:

$$T = T_{ON} + T_{OFF}$$

where T_{ON} is the time during which the output is HIGH, and T_{OFF} is the time during which the output is LOW.

The ratio between T_{ON} and the total period T is called the **duty cycle (D)**:

$$D = \frac{T_{ON}}{T}$$

For this project, we want a duty cycle close to **50%**.

For a standard NE555 astable configuration:

$$T_{ON} = 0.69(R1 + R2)C \quad (1)$$

$$T_{OFF} = 0.69 \times R2 \times C \quad (2)$$

Therefore:

$$T = 0.69(R1 + 2R2)C \quad (3)$$

Since:

$$f = \frac{1}{T}$$

We want:

$$f = 10 \text{ Hz}$$

which gives:

$$T = \frac{1}{10} = 0.1 \text{ s}$$

Let's choose convenient values for the capacitor and R1.

$$C = 10 \mu F = 0.00001 F$$

$$R1 = 1 K = 1000 \Omega$$

We chose 1 k Ω for R1 to obtain a duty cycle close to 50%.

Then, let's calculate R2.

From Equation (1), we get:

$$2R2 = \frac{T}{0.69C1} - R1$$

$$R2 = \left(\frac{T}{0.69C1} - R1 \right) / 2$$

$$R2 = \left(\frac{0.1}{0.69 \times 0.00001} - R1 \right) / 2$$

$$= 6746.37 \Omega$$

$$= 6.7\text{k}\Omega$$

We can use the standard **6.8 k Ω** value.

So our 10 Hz oscillator will use approximately:

- **R1 = 1 k Ω**
- **R2 = 6.8 k Ω**
- **C = 10 μ F**

This gives a frequency of approximately **9.9 Hz** with a duty cycle of about **53%**, which is close to our target of 50%.

2.2 – 38 kHz Astable Oscillator

The second NE555 generates the **38 kHz carrier** that drives the IR LED.

We use the same astable configuration:

$$[T=0.69(R_1+2R_2)C]$$

and:

We want:

$$f = 38 \text{ kHz} = 38000 \text{ Hz}$$

$$T = \frac{1}{38000} = 0.000026\text{s}$$

For a 38 kHz carrier:

Let's choose convenient values for the capacitor C2 and R5.

$$\mathbf{C = 1nF = 10^{-9} \text{ F}}$$

$$\mathbf{R5 = 1k = 1000\Omega}$$

Then, let's calculate R6:

$$R6 = \left(\frac{T}{0.69C1} - R5 \right) / 2$$

$$R6 = \left(\frac{0.00002}{0.69C2} - 1000 \right) / 2$$

$$= 18840\Omega$$

$$= 18.8k\Omega$$

So, we are going to use a 20 kΩ potentiometer, which is close to the calculated 18.5 kΩ value, to adjust the frequency.

So our 10 Hz oscillator will use approximately:

- **R5 = 1 kΩ**
- **R6 = 20 kΩ**
- **C = 10 nF**

Step 3: Dimension the IR LED Current-Limiting Resistor

Our emitter uses **one IR LED**. We want the LED to emit a strong infrared signal in order to increase the range of the emitter.

To limit the current, we use a **series resistor** with the LED.

Using Kirchhoff's Voltage Law:

$$V_{CC} - V_{LED} - RI_{LED} = 0$$

The target LED current is 20mA(0.02A):

$$R = \frac{V_{CC} - V_{LED}}{I_{LED}}$$

$$= \frac{12 - 2}{0,02}$$

$$= \frac{12 - 2}{0,02}$$

$$= 500 \Omega$$

we are going to use a 490 Ω resistor, which is close to the calculated value.

We also need to check the power dissipated by the resistor:

$$P_R = R I^2$$

$$P_R = 490 \times 0.024^2 = 0.282W$$

A **1/2 W (0.5 W)** resistor is therefore a suitable choice.

Final values

- **1 × IR LED**
- **1 × 490 Ω resistor**
- **Approximately 20mA LED current**
- **1/2 W resistor**

Step 4: Build the Circuit

Components

- 2 × NE555 timer ICs
- 1 × IR LED
- 2 × 1 k Ω resistor
- 1 × 6.8 k Ω resistor
- 1 × 20k Ω resistor
- 1 × 10 μ F capacitor
- 1 × 1 nF capacitor
- 1 × 4900 Ω , 1/2 W resistor
- 12 V DC power supply
- Breadboard and jumper wires

The complete circuit schematic and the final prototype are shown below.

Step 5: Test the Circuit

1. Check the Oscillation Frequencies

Connect the outputs of the two oscillators to an **oscilloscope** and verify their frequencies:

- The first oscillator should generate approximately **10 Hz**.
- The second oscillator should generate a carrier close to **38 kHz**.
- Adjust the potentiometer of the second oscillator until the carrier frequency is precisely **38 kHz**.

The 38 kHz carrier should appear as **short bursts controlled by the first oscillator**, with pauses between them.

2. Check the IR LEDs

Point the IR LEDs toward a **smartphone camera**. Most phone cameras can detect infrared light, allowing you to see the LEDs flashing when the emitter is operating.

The LEDs should appear to **flash periodically**, confirming that the 38 kHz carrier is being transmitted in bursts.

What's Next?

In the next Instructables, we will build different **IR receiver circuits for specific applications**, all using the same IR emitter developed in this project:

- **Automatic Handwashing System** — an IR beam barrier to detect the presence of a user's hands and activate the water flow automatically.
- **Presence Detector** — an IR-based system for detecting when an object or person crosses the beam.
- **Remote-Controlled Lamp with a CD4017** — an IR receiver combined with a CD4017 to control a lamp remotely.

The advantage is that **the same 38 kHz IR emitter can be reused in all these projects**. Only the receiver and the processing circuit will change depending on the application.